Robust Online POMDP/MDP MCTS Planning with Anytime Deterministic Guarantees

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Abstract

Despite significant progress in robust planning, existing robust POMDP solvers remain offline, while online robust methods are restricted to fully observable MDPs and provide only probabilistic guarantees. We address this gap by introducing the first online anytime planner for robust POMDPs with deterministic guarantees. Our planner operates on a projected problem over sampled states and observations while deterministically bounding the probability mass of unsampled trajectories, yielding an anytime certificate on the approximation error. For any fixed policy, this certificate deterministically bounds its robust value under both rectangular and non-rectangular uncertainty sets. For rectangular uncertainty sets, applying the robust Bellman optimality operator further yields a deterministic upper bound on the optimal robust value. Together, these bounds guide online search and provide a certified stopping criterion for anytime planning, enabling early termination while guaranteeing optimality with respect to the original robust POMDP rather than its projected approximation. Building on these results, we present R-MCTS, the first online Monte Carlo tree search framework for robust POMDPs. We instantiate two variants: one using UCB exploration, which converges to the robust value of its induced tree policy, and another guided by our certified upper bound, which converges to the optimal robust value.With minor adjustments, our results also apply to robust MDPs. Experimental results validate our theoretical guarantees, demonstrating that the certified intervals contain the true robust value and tighten as the sampled support grows.

Introduction

contributions:

1.A deterministic, anytime bound on the gap between the projected robust value of a fixed policy and its true robust value, $|V_0^{\pi,rob} - \widehat{V}_0^{\pi,rob}| \leq \epsilon_0$, computable at no additional complexity from the sampled tree; it holds at the root for any belief and, under exact belief updates, at every node.

2.Optimal-value upper bound. A deterministic robust upper bound on the optimal robust value, $V_{t,rect}^{*,rob}(b_t) \leq RUDB^{\pi}(b_t)$, obtained by applying the robust Bellman optimality operator to the certified upper bound.

3. RUDB exploration. An exploration criterion that uses this upper bound to guide search, made valid at every node by decoupling belief update from robust backup — exact filtering with the nominal transition for validity, projected optimization for tractability — so the optimal-value certificate is available per-node without full-set optimization cost.

4. Both bounds are asymptotically exact: for any support-expanding online strategy $\widehat{\mathcal{T}} \to \mathcal{T}$, the projected values converge to the true robust value $V^{\pi,rob}$ and, under the RUDB policy, to the optimal robust value $V_{rect}^{*,rob}$ The convergence rate is governed by the support-expansion rate, decoupling the guarantee from the specific solver.

5. R-MCTS framework and two solvers. A robust online planning framework that adds the above certificates to sampling-based search, instantiated as two solvers: (i) a POMCP-style planner with UCB exploration and robust projected backups, which inherits the fixed-policy certificate and converges to the robust value of the tree-policy; and (ii) an RUDB-exploration planner with exact (nominal) belief updates, which additionally attains the optimal-value guarantee and converges to the optimal robust value.

6. Certified early stopping and pruning. The two-sided bracket certifies action orderings during search: provably-suboptimal subtrees are pruned, and planning stops early once the optimal action is certified — rather than exhausting the sampling budget.

7. Post-hoc certification: the bound depends only on the evaluated policy and not on how it was produced, it certifies the robust value of any policy or planner, irrespective of how it was obtained.

8. A projection method that reduces a full RPOMDP to a tractable problem on the sampled support, under which robust backups optimize over projected sets rather than the full high-dimensional ones — and which provably preserves the worst-case value on that support.

Preliminaries

A robust POMDP accounts for model uncertainty by replacing the single transition and observation models with a set of admissible models. We consider a finite-horizon non-stationary RPOMDP $\mathcal{M}^{\text{rob}} = (\mathcal{S}, \mathcal{A}, \mathcal{Z}, \mathcal{P}^T_{0:H-1}, \mathcal{P}^Z_{1:H}, b_0, r, H)$, where $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{Z}$ are the finite state, action, and observation spaces. The sets $\mathcal{P}^T_{0:H-1}, \mathcal{P}^Z_{1:H}$ contain admissible non-stationary transition and observation model sequences $P^T_{0:H-1}, P^Z_{1:H}$, over the horizon H. The information available to the agent is summarized by the history of actions and observations up to time t which is denoted by $h_t \triangleq \{b_0, a_{0:t-1}, z_{1:t}\}$. This history induces a posterior distribution over the current hidden state, called the belief, defined by $b_t(s) \triangleq \Pr(s_t = s \mid h_t)$, b_0 being the initial prior belief. At each time step t after taking action a_t and observing z_{t+1}, the belief is updated with the Bayesian filter $b_{t+1}(s_{t+1} \mid h_t, a_t, z_{t+1}) = \eta_t \cdot P_Z(z_{t+1} \mid s_{t+1}) \sum_{s_t \in \mathcal{S}} P_T(s_{t+1} \mid s_t, a_t) b_t(s_t)$, where η_t is the normalizing constant. The reward function $r : \mathcal{B}(\mathcal{S}) \times \mathcal{A} \to [r_{min}, r_{max}]$ is assumed to be bounded and state-dependent, and is defined as $r(b, a) = \mathrm{E}_{s\sim b}[r(s, a)]$. A policy specifies how actions are selected based on the current belief. For a finite-horizon problem, a policy is a sequence $\pi = \{\pi_t\}_{t=0}^H$, where each π_t maps a belief b_t to a distribution over actions: $\pi_t(a_t \mid b_t) \triangleq \Pr(a_t = a \mid b_t)$.

Given a sequence of nominal models $\mathring{P}^T_{0:H-1}, \mathring{P}^Z_{1:H}$, such that the kernels at time step k are $\mathring{P}^T_k : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, $\mathring{P}^Z_k : \mathcal{S} \to \Delta(\mathcal{Z})$, the non-rectangular uncertainty sets over the planning horizon H are defined as,

$$\mathcal{P}^T_{0:H-1} = \{P^T_{0:H-1} : \; D_T\left(P^T_{0:H-1} || \mathring{P}^T_{0:H-1}\right) \leq \rho^T\} \quad (1)$$

$$\mathcal{P}^Z_{1:H}, = \{P^Z_{1:H} : \; D_Z\left(P^Z_{1:H} || \mathring{P}^Z_{1:H}\right) \leq \rho^Z\}, \quad (2)$$

where D_T, D_Z are some distance metrics, and ρ_T and ρ_Z are given uncertainty radiuses. The robust value function evaluates π under the worst-case admissible model, and is defined as:

$$V^{\pi,\text{rob}}_t(b_t) = \inf_{\substack{P^T_{t:H-1} \in \mathcal{P}^T_{t:H-1} \\ P^Z_{t+1:H} \in \mathcal{P}^Z_{t+1:H}}} V^\pi_t(b_t; P^T_{t:H-1}, P^Z_{t+1:H}), \quad (3)$$

where $V^\pi_t(b_t; P^T_{t:H-1}, P^Z_{t+1:H})$ is the value function given a sequence of models. To reduce clutter, when clear from context, we omit the explicit dependence on $P^T_{t:H-1}$ and $P^Z_{t+1:H}$, and write $V^\pi_t(b_t; P^T_{t:H-1}, P^Z_{t+1:H})$ as

$$V^\pi_t(b_t) = r(b_t, \pi_t) + \mathbb{E}_{z_{t+1:H}} \sum_{j=t+1}^{H-1} r(b_j, \pi_j). \quad (4)$$

The non-rectangular formulation in (1)-(3), forces the adversary to choose the model sequence jointly, so the admissibility of the model at one time may depend on choices at others. This coupling prevents a dynamic-programming decomposition of the Bellman backup, motivating the rectangular relaxation. The rectangular sets at time step t are defined as: [this paragraph can still be clarified and reduced]

$$\mathcal{P}^{T,rect}_t = \bigotimes_{\substack{s\in\mathcal{S} \\ a\in\mathcal{A}}} \mathcal{P}^T_t(s,a), \quad \mathcal{P}^{Z,rect}_{t+1} = \bigotimes_{s\in\mathcal{S}} \mathcal{P}^Z_{t+1}(s), \quad (5)$$

$$\mathcal{P}^T_t(s,a) = \left\{P^T_t(\cdot|s,a) \colon D\left(P^T_t(\cdot|s,a) || \mathring{P}^T_t(\cdot|s,a)\right) \leq \rho^T_{s,a}\right\},$$

$$\mathcal{P}^Z_{t+1}(s) = \{P^Z_{t+1}(\cdot|s) \colon D\left(P^Z_{t+1}(\cdot|s) || \mathring{P}^Z_t(\cdot|s)\right) \leq \rho^Z_s\right\},$$

Rectangularity is imposed at four levels. First, nature may choose a different worst-case model at each stage independently of its choices made at other stages. Second, within each stage, transition and observation uncertainty are independent. Third, row-rectangular, every transition/observation row representing the conditional distribution $P^T_t(\cdot|s_t, a_t) \in \Delta\mathcal{S}$, $P^Z_{t+1}(\cdot|s_{t+1}) \in \Delta\mathcal{Z}$, is chosen independently. Fourth, history-rectangular, different nodes of the same step in the tree are independent. This represents an adversary that may adapt its models choices locally rather than committing in advance to a fixed sequence of models.

Adapting (3) to the rectangular setting, we denote the robust rectangular value function, that admits robust DP:

$$V^{\pi,\text{rob}}_{t,\text{rect}}(b_t) = r(b_t, \pi_t) + \inf_{\substack{P^T_t \in \mathcal{P}^T_t \\ P^Z_{t+1} \in \mathcal{P}^Z_{t+1}}} \mathbb{E}^{P^T_t, P^Z_{t+1}}_{z_{t+1}} \left[V^{\pi,\text{rob}}_{t+1,\text{rect}}(b_{t+1})\right]. \quad (6)$$

The rectangular relaxation comes at a cost of a more conservative robust value than the non-rectangular robust value. Since $\mathcal{P}_{t:H} \subseteq \mathcal{P}^{rect}_{t:H}$, we get $V^{\pi,\text{rob}}_{t,\text{rect}}(b_t) \leq V^{\pi,\text{rob}}_t(b_t)$.

[might need to add a section in the Preliminaries on MCTS - let's decide later]

Certified Anytime Robust Online Solver

Computing the exact optimal robust value is typically infeasible in online POMDP planning, since the belief tree grows exponentially and each robust backup requires optimization over transition and observation uncertainty sets. We therefore plan over a projected belief tree containing only the sampled states and observations at each history/action node. The planner computes robust values on this projected problem, while the probability mass of trajectories leaving the sampled support is not ignored: it is bounded deterministically and used as an anytime certificate on the approximation error. We apply bellman optimality over the certified robust upper bounds which bounds the optimal rectangular value function. [not sure if we should also mention here that as sampled support increase the bounds converge to the optimal rect value]

For some time step t in the horizon, the state-support is the sampled set $\widehat{\mathcal{S}}(h^-_t) \subseteq \mathcal{S}$, the observation support is $\widehat{\mathcal{Z}}(h_t) \subseteq \mathcal{Z}$, and the action support is $\widehat{\mathcal{A}}(h^-_t) \subseteq \mathcal{A}$. Let us define the set of tree-trajectories, consistent with the sampled states/observations/action support: [why do we not have also a subspace of the Action space?][why do we need $h_{t:H}$? can just use h_H, and then h_t will be a subsequence of it for any

$t \in [1, H]]$

$$\widehat{\mathcal{T}}_{t:H}(h_H) = \widehat{\mathcal{S}}(h_t^-)\times\widehat{\mathcal{A}}(h_t^-)\times\widehat{\mathcal{S}}(h_{t+1}^-)\times\widehat{\mathcal{Z}}(h_{t+1})\times\cdots \times\widehat{\mathcal{A}}(h_H)\times\widehat{\mathcal{S}}(h_H^-)\times\widehat{\mathcal{Z}}(h_H), \tag{7}$$

such that a single trajectory sample is denoted by $\tau_{t:H} \in \widehat{\mathcal{T}}_{t:H}(h_{t:H})$. Important nuance, if for any time step $k \in [t, H]$, $s_k \notin \widehat{\mathcal{S}}(h_k^-)$ or $z_k \notin \widehat{\mathcal{Z}}(h_k)$, then $\tau_{t:H} \notin \widehat{\mathcal{T}}_{t:H}$. [revisit paragraph] The sampled trajectory set identifies the branches represented in the planning tree, we now define the projected transition and observation models $\widehat{P}_t^T$, $\widehat{P}_{t+1}^Z$ to be sub-probabilities, defined only on their relative state/observation support:

$$\widehat{P}_t^T(\cdot|s_t, a_t; h_{t+1}^-) \triangleq P_t^T(\cdot|s_t, a_t),\ \forall s_{t+1}\in\widehat{\mathcal{S}}(h_{t+1}^-), \tag{8}$$

$$\widehat{P}_{t+1}^Z(\cdot|s_{t+1}; h_{t+1}) \triangleq P_{t+1}^Z(\cdot|s_{t+1}), \forall z_{t+1}\in\widehat{\mathcal{Z}}(h_{t+1}), \tag{9}$$

where P_t^T, P_{t+1}^Z belong to the full uncertainty sets, and $\widehat{P}_t^T \in \Delta\widehat{\mathcal{S}}(h_{t+1}^-)$, and $\widehat{P}_{t+1}^Z \in \Delta\widehat{\mathcal{Z}}(h_{t+1})$, meaning the projected models are over a much lower dimensionality, so robust backups optimize over $\mathbb{R}^{|\widehat{\mathcal{S}}|}$ rather then $\mathbb{R}^{|\mathcal{S}|}$. NOT SURE I NEED THIS:

The projected prior belief $\widehat{b}_0$, is the belief prior b_0 of the RPOMDP restricted to the state-support at the root node

$$\widehat{b}_0(\cdot) \triangleq \begin{cases} b_0(s_0), & s_0 \in \widehat{\mathcal{S}}(h_0^-), \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The posterior belief $\widehat{b}_t$ is updated using the projected models, $\widehat{b}_t(s_t; \widehat{\mathcal{S}}(h_t)) = \widehat{P}_{t+1}^Z(z_t|s_t)\sum_{s_{t-1}\in\widehat{\mathcal{S}}(h_t^-)} \widehat{P}_t^T(s_t|s_{t-1}, a_{t-1})\widehat{b}(s_{t-1})$. Thus, the unnormalized projected models induce an unnormalized posterior belief.

Using these we define the projected value function:

$$\widehat{V}_t^\pi(\widehat{b}_t) \triangleq r(\widehat{b}_t, \pi_t) + \widehat{\mathbb{E}}_{z_{t+1}\sim\widehat{\Pr}(\cdot|\widehat{b}_t,\pi_t)}\left[\widehat{V}_{t+1}^\pi(\widehat{b}_{t+1})\right],$$

$$= \sum_{j=t}^{H-1}\left[\sum_{\tau_{t:j}\in\widehat{\mathcal{T}}_{t:j}}\prod_{k=t}^{j-1}\widehat{P}_k^T(s_{k+1}|\pi_k, s_k)\widehat{P}_{k+1}^Z(z_{k+1}|s_{k+1}) \times\widehat{b}(s_t)r(s_j, a_j)\right]. \tag{11}$$

Projected Uncertainty Set

[we'll need to adjust later the opening here] Our online planner is robust with respect to sampled subsets of the planning tree, which implies that the uncertainty around the projected models significantly differs from the uncertainty around the nominal models of the RPOMDP.

We introduce a new concept we call the *projected uncertainty set*. This set is defined as the set of **all** valid sub-probability vectors such that **each** vector can be extended to a full probability distribution that lies within the original uncertainty set, namely:

$$\widehat{\mathcal{P}}_{t:H-1}^T(h_{H-1}) = \left\{\widehat{P}_{t:H-1}^T : \exists P_{t:H-1}^{T,out} \text{ s.t } \left(\widehat{P}_{t:H-1}^T \cup P_{t:H-1}^{T,out}\right) \in \mathcal{P}_{t:H-1}^T\right\} \tag{12}$$

$$\widehat{\mathcal{P}}_{t+1:H}^Z(h_H) = \left\{\widehat{P}_{t+1:H}^Z : \exists P_{t+1:H}^{Z,out} \text{ s.t } \left(\widehat{P}_{t+1:H}^Z \cup P_{t+1:H}^{Z,out}\right) \in \mathcal{P}_{t+1:H}^Z\right\} \tag{13}$$

where for each time step t in the horizon, $P_t^{T,out} \in \mathbb{R}^{|\mathcal{S}^{\text{out}}(h_{t+1}^-)|}$, $P_{t+1}^{Z,out} \in \mathbb{R}^{|\mathcal{Z}^{\text{out}}(h_{t+1})|}$ are the complementary models to their project counterparts, and their dimension is the size of the unsampled-support. These projected uncertainty sets catch the original uncertainty of the RPOMDP with respect to the sampled support.

[== stopped here ==]

Projected Robust POMDP

The projected RPOMDP, is defined as the tuple $\widehat{\mathcal{M}}^{\text{rob}} = (\widehat{\mathcal{S}}, \widehat{\mathcal{A}}, \widehat{\mathcal{Z}}, \widehat{\mathcal{P}}_{t:H-1}^T, \widehat{\mathcal{P}}_{t+1:H}^Z, b_0, r, H)$. Given some policy π, the projected [estimated? need to decide] robust value function is defined as the worst case models choices over the projected uncertainty set:

$$\widehat{V}_t^{\pi,\text{rob}}(\widehat{b}_t) = \inf_{\substack{\widehat{P}_{t:H-1}^T\in\widehat{\mathcal{P}}_{t:H-1}^T \\ \widehat{P}_{t+1:H}^Z\in\widehat{\mathcal{P}}_{t+1:H}^Z}} \widehat{V}_t^\pi(\widehat{b}_t; \widehat{P}_{t:H-1}^T, \widehat{P}_{t+1:H}^Z) \tag{14}$$

(14) is the same objective (3) the agent seeks to maximize, but evaluated only over valid trajectories represented in the sampled tree.

Anytime Deterministic Guarantees

This section analyzes the performance guarantees of our approach, relating the approximated robust value function (14) to the theoretical robust value (3):

[Thm1 is valid for any uncertainty set, right? Not necessarily even a ball uncertainty set (1). Maybe we should have a notation for a general non-rectangular uncertainty set]

Theorem 1. *Let b_0 be the belief at time $t = 0$, $V_0^{\pi,\text{rob}}(b_0)$, $\widehat{V}_0^{\pi,\text{rob}}(b_0)$ be the theoretical/projected (14) robust value function that is achieved by following the policy π, and $\widehat{\mathcal{P}}_{0:H-1}^T$ and $\widehat{\mathcal{P}}_{1:H}^Z$ be the projected, non-rectangular transition and observation uncertainty sets* (12)-(13). *For any policy π and horizon H, the difference between the theoretical and the projected robust value functions is bounded deterministically by:*

$$\left|V_0^{\pi,\text{rob}}(b_0) - \widehat{V}_0^{\pi,\text{rob}}(b_0)\right| \leq \epsilon_0^\pi(b_0) \triangleq \tag{15}$$

$$R_{\max}\left(H-1-\inf_{\substack{\widehat{P}_{0:H-1}^T\in\widehat{\mathcal{P}}_{0:H-1}^T \\ \widehat{P}_{1:H}^Z\in\widehat{\mathcal{P}}_{1:H}^Z}}\sum_{j=1}^{H-1}\sum_{\tau_{0:j}\in\widehat{\mathcal{T}}_{0:j}}\widehat{\Pr}(\tau_{0:j})\right) \tag{16}$$

where the trajectory likelihood $\widehat{\Pr}(\tau_{0:j}) = \widehat{b}_0(s_0)\prod_{k=0}^{j-1}\widehat{P}_k^T(s_{k+1}|\pi_k, s_k)\widehat{P}_{k+1}^Z(z_{k+1}|s_{k+1})$ is used to reduce clutter.

Corollary 1. *The bound of Theorem 1 holds for every node h_t of the tree, given that along the trajectory to h_t ,each previous node's belief was updated over the full state space (exact belief updates), namely:*

$$\left|V_t^{\pi,\mathrm{rob}}(b_t) - \widehat{V}_t^{\pi,\mathrm{rob}}(b_t)\right| \leq \epsilon_t^{\pi}(b_t) \quad (17)$$

$$\epsilon_t^{\pi}(b_t) \triangleq R_{max}\left(H-t-1- \inf_{\substack{\widehat{P}^T_{t:H-1}\in\widehat{\mathcal{P}}^T_{t:H-1}\\ \widehat{P}^Z_{t+1:H}\in\widehat{\mathcal{P}}^Z_{t+1:H}}} \sum_{j=t+1}^{H-1} \sum_{\tau_{t:j}\in\widehat{\mathcal{T}}_{t:j}} \widehat{\Pr}(\tau_{t:j})\right) \quad (18)$$

add the definition of $\epsilon_{t,rect}$

Note: the bound in (18) is still a function of the projected models $P^T_{t:H-1}, P^Z_{t+1:H}$ resulting in an optimization over a much lower dimensionality! We thus pay additional computation only in the belief-update, while the optimizations remain cheap.

Corollary 2. *For any policy π, as the sampled trajectories set increases $\widehat{\mathcal{T}} \to \mathcal{T}$ the projected robust value function* (14) *convergences to the theoretical robust value function* (3)*:*

$$\widehat{V}_0^{\pi,\mathrm{rob}}(\widehat{b}_0) \xrightarrow{\widehat{\mathcal{T}}\to\mathcal{T}} V_0^{\pi,\mathrm{rob}}(b_0). \quad (19)$$

Crucially Theorem 1 makes no assumption on the structure of the uncertainty sets, and thus holds precisely for rectangular uncertainty sets.

Instantiating it in the rectangular setting bounds the rectangular robust value at the root, $\left|V_{0,\mathrm{rect}}^{\pi,\mathrm{rob}}(b_0) - \widehat{V}_{0,\mathrm{rect}}^{\pi,\mathrm{rob}}(\widehat{b}_0)\right| \leq \epsilon^{\pi}_{0,rect}(\widehat{b}_0)$, which is the setting we adopt in the sequel: rectangularity admits a dynamic-programming decomposition of the robust Bellman backup, while the bound, computable without extra complexity, provides an anytime deterministic certificate of policy quality.

By Corollary 2, this certificate is asymptotically exact: the interval collapses [converges] onto the true robust value as the sampled support grows $\widehat{\mathcal{T}} \to \mathcal{T}$. The rate of collapse [convergence rate] is governed by how quickly the algorithm expands its support, thus the guarantee tightens monotonically with planning effort, making it well suited to online planning. [you mention here: monotonicity, asymptotic convergence, and convergence rate - consider formulating explicitly as properties]

Two uses follow. [state each in a separate paragraph, can use bold text in the beginning] First, the bound yields an early-stopping criterion. It brackets each action value in the interval $[\widehat{Q}^{\pi,rob}_{rect} - \epsilon^{\pi}_{rect}(\widehat{b},a),\ \widehat{Q}^{\pi,rob}_{rect} + \epsilon^{\pi}_{rect}(\widehat{b},a)]$, where $\epsilon^{\pi}_{rect}(\widehat{b},a)$ is the action-value error defined in (**??**). When two actions' intervals do not overlap, their ordering is certified: an action whose upper bound falls below another's lower bound is provably suboptimal, and its sub-tree can safely be pruned from further search, while an action whose lower bound exceeds every other upper bound is certified optimal. Our algorithm applies this test at the root, terminating early once the root action is certified optimal rather than exhausting the budget. Under exact belief updates (Corollary 1) the same test applies at every node, pruning certified-suboptimal sub-trees and eliminating redundant search. [the first one is related to the next section] Second, the bound depends on the evaluated policy alone, not on its origin; it therefore yields post-hoc guarantees on the robust value of any policy or planning algorithm, irrespective of how it was obtained.

Post-hoc guarantees cccc [todo]

Early stopping and suboptimal pruning cccc [todo]

Optimality Guarantees

Theorem 1 bounds the projected value of a fixed policy π from above and below. We now leverage this to bound the *optimal* robust value $V^{*,\mathrm{rob}}_{t,rect}$ by applying the Bellman optimality operator to the robust upper deterministic bound (RUDB) we define as: [clarify this is in the setting of exact belief updates and a rectangular uncertainty set]

$$RUDB^{\pi}(b_t,a_t) \triangleq r(b_t,a_t) + \inf_{\substack{\widehat{P}^T_t\in\widehat{\mathcal{P}}^T_t\\ \widehat{P}^Z_{t+1}\in\widehat{\mathcal{P}}^Z_{t+1}}} \left\{ \widehat{\mathbb{E}}^{\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[\widehat{V}^{\pi,RUDB}_{t+1,rect}(b_{t+1})\right] + \epsilon^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{t,rect}(b_t,a_t)\right\}, \quad (20)$$

where the action-value error is defined by fixing action a_t and following π in the time steps $j\in[t+1,H]$ that follow:

$$\epsilon^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{t,rect}(b_t,a_t) \triangleq R_{max}(H-t-1- \sum_{s_t\in\widehat{\mathcal{S}}_t} b_t(s_t)\delta_t^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}(s_t,a_t), \quad (21)$$

$$\delta_t^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}(s_t,a_t) = \sum_{\substack{z_{t+1}\in\widehat{\mathcal{Z}}_{t+1}\\ s_{t+1}\in\widehat{\mathcal{S}}_{t+1}}} \widehat{P}^T_t(s_{t+1}|a_t,s_t) \times \widehat{P}^Z_{t+1}(z_{t+1}|s_{t+1})\left[1+\delta^{\pi}_{t+1}(s_{t+1})\right], \quad (22)$$

The RUDB superscript denotes the current-step model $\widehat{P}^{*T}_t, \widehat{P}^{*Z}_{t+1}$ that jointly minimize (20); the value term $\widehat{Q}^{\pi,RUDB}_{t,rect}$ and the error term $\epsilon^{\pi,RUDB}_{t,rect}$ are both evaluated at these shared minimizers, rather than optimized independently. This coupling is what allows the bound to be assembled from a single robust backup.

Theorem 2. *Let b_t be the belief at time step t, $\mathring{P}^T_t$ be the nominal transition model that is used to update the posterior belief $b_{t+1}(s_{t+1}|h^-_t, z_{t+1}; \mathring{P}^T_t) = \Pr(s_{t+1}|h_t)$, called the nominal-belief at time step $t+1$. The optimal robust value function at time step t over rectangular uncertainty sets, is upper bounded by:*

$$V^{*,\mathrm{rob}}_{t,rect}(b_t) \leq RUDB^{\pi}(b_t) \triangleq \max_{a_t\in\mathcal{A}}\left\{RUDB^{\pi}(b_t,a_t)\right\}, \quad (23)$$

such that the policy π comes from applying Bellman optimality operator over $RUDB^{\pi}(b_t,a_t)$:

$$\pi_t(b_t) = \arg\max_{a_t\in\mathcal{A}_t}\left\{\widehat{Q}^{\pi,RUDB}_{t,rect}(b_t,a_t) + \epsilon^{\pi,RUDB}_{t,rect}(b_t,a_t)\right\}, \quad (24)$$

A subtlety underlies the use of Theorem 2 as an exploration criterion. For the RUDB to guide search, the error must be a valid bound at every node of the tree, not only the root—which requires the posterior belief at each node to be exact. We therefore update beliefs with the full nominal transition model $\mathring{P}_t^T$ yielding an exact posterior at every history node. Crucially, this exactness is confined to the belief update: the robust backups still optimize over the projected uncertainty sets $\widehat{\mathcal{P}}_t^T, \widehat{\mathcal{P}}_{t+1}^Z$ rather than the full sets. We thus pay additional computation only in the filter, while the backups—which carry the robustness—remain cheap, as each infimum ranges over the sampled support alone. [where does the sampled state support come from?]

This yields two practical consequences. First, under exact belief updates, the two-sided bracket certifies action orderings at every node, not only the root: the pruning and early-stopping criteria of the previous section thus apply throughout the tree, eliminating redundant search.

Finally, as the sampled support expands, the guarantee becomes exact:

Corollary 3. *Let b_t be the belief as time step t, and π^{RUDB} be the RUDB policy* (24)*. As the sampled trajectories set increases $\widehat{\mathcal{T}} \to \mathcal{T}$, the projected robust value function [need to define it somewhere] convergences to the optimal robust value function over rectangular uncertainty sets, namely:*

$$\widehat{V}_{t,rect}^{\pi^{RUDB},\mathrm{rob}}(b_t) \xrightarrow{\widehat{\mathcal{T}} \to \mathcal{T}} V_{t,rect}^{*,\mathrm{rob}}(b_t) \tag{25}$$

[need to add the proof in supplementary]

Supplementary

Proof of Theorem 1

Essentials lemmas and definitions for the proof:

1. The set of all trajectories that is consistent with the theoretical planning tree:

$$\mathcal{T}_{t:H} = S_t \times A \times S_{t+1} \times Z_{t+1} \times \cdots \times A \times S_H \times Z_H \tag{26}$$

2. The sets of trajectories where at least 1 state/observation sample is not consistent with the sampled planning tree:

$$\mathcal{T}^{\text{out}}_{t:H} = \mathcal{T}_{t:H} / \widehat{\mathcal{T}}_{t:H} \tag{27}$$

3. The restriction mapping $\mathcal{R}$:

$$\mathcal{R}^T_{t:k-1}(P^T_{t:k-1}; h_{t:k-1}) = \begin{cases} P^T_{t:k-1}, & s_{t:k-1} \in \widehat{\mathcal{S}}(h_{t:k-1}), \\ 0, & \text{otherwise.} \end{cases} \tag{28}$$

$$\mathcal{R}^Z_{t+1:k}(P^Z_{t+1:k}; h_{t+1:k}) = \begin{cases} P^Z_{t+1:k}, & z_{t+1:k} \in \widehat{\mathcal{Z}}(h_{t+1:k}), \\ 0, & \text{otherwise.} \end{cases} \tag{29}$$

Lemma 1 (restated). *Let H be the horizon of the POMDP, and π be the policy. the theoretical/projected value function V^π_t, $\widehat{V}^\pi_t$ at time step t equals:*

$$V^\pi_t(b_t) = \sum_{j=t}^{H-1} \sum_{z_{t+1:j}} \sum_{s_{t:j}} b_t(s_t) \prod_{k=t}^{j-1} P^T_k(s_{k+1} \mid s_k, \pi_k) P^Z_{k+1}(z_{k+1} \mid s_{k+1}) r\big(s_j, \pi_j(h_j)\big), \tag{30}$$

$$\widehat{V}^\pi_t(b_t) = \sum_{s_t} b_t(s_t) r\big(s_t, \pi_t(h_t)\big) + \sum_{j=t+1}^{H-1} \sum_{\substack{z_{t+1:j} \in \widehat{\mathcal{Z}}(h_{t+1:j}) \\ s_{t:j} \in \widehat{\mathcal{S}}(h_{t:j})}} \widehat{b}_t(s_t) \prod_{k=t}^{j-1} \widehat{P}^T_k(s_{k+1}|s_k, \pi_k) \widehat{P}^Z_{k+1}(z_{k+1}|s_{k+1}) r\big(s_j, \pi_j(h_j)\big). \tag{31}$$

Lemma 2 (restated). *for any 2 functions $f(\theta)$ and $\widehat{f}(\theta)$, the following holds:*

$$\left| \min_\theta f(\theta) - \min_\theta \hat{f}(\theta) \right| \leq \max_\theta \left| f(\theta) - \hat{f}(\theta) \right| \tag{32}$$

Lemma 3 (restated). *Let $\mathcal{P}$ denote some uncertainty set, and $\widehat{\mathcal{P}}$ denote its projected version. Let $\mathcal{R}$ be the restriction map that takes a full probability distribution $P \in \mathcal{P}$ and retains only the entries associated with the sampled support:*

$$\mathcal{R}(P; \widehat{\mathcal{S}}) = \begin{cases} P(s), & s \in \widehat{\mathcal{S}}, \\ 0, & \textit{otherwise}. \end{cases} \tag{33}$$

Then minimizing some function $f : \widehat{P} \to \mathbb{R}$ over $\widehat{\mathcal{P}}$ is the same as minimizing $f(\mathcal{R}(P))$ over the full set $\mathcal{P}$, namely:

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) = \inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}). \tag{34}$$

Theorem 1 (restated). *Let b_0 be the belief at time $t = 0$, $V^{\pi,\text{rob}}_0(b_0)$, $\widehat{V}^{\pi,\text{rob}}_0(b_0)$ be the theoretical/projected (14) robust value function that is achieved by following the policy π, and $\widehat{\mathcal{P}}^T_{0:H-1}$ and $\widehat{\mathcal{P}}^Z_{1:H}$ be the projected, non-rectangular transition and observation uncertainty sets* (12)-(13)*. For any policy π and horizon H, the difference between the theoretical and the projected robust value functions is bounded deterministically by:*

$$\left| V^{\pi,\text{rob}}_0(b_0) - \widehat{V}^{\pi,\text{rob}}_0(b_0) \right| \;\leq\; \epsilon^\pi_0(b_0) \triangleq \tag{35}$$

$$R_{max} \left(H - 1 - \inf_{\substack{\widehat{P}^T_{0:H-1} \in \widehat{\mathcal{P}}^T_{0:H-1} \\ \widehat{P}^Z_{1:H} \in \widehat{\mathcal{P}}^Z_{1:H}}} \sum_{j=1}^{H-1} \sum_{\substack{z_{1:j} \in \widehat{\mathcal{Z}}(h_{1:j}) \\ s_{0:j} \in \widehat{\mathcal{S}}(h_{0:j})}} \widehat{b}_0(s_0) \prod_{k=0}^{j-1} \widehat{P}^T_k(s_{k+1} \mid \pi_k, s_k)\, \widehat{P}^Z_{k+1}(z_{k+1} \mid s_{k+1}) \right)$$

Proof.
The robust non-rectangular value function is defined as:

$$V_0^{\pi,\mathrm{rob}}(b_0) \triangleq \inf_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} V_0^{\pi}\big(b_0;\, P^T_{0:H-1}, P^Z_{1:H}\big)\,, \tag{36}$$

and the projected robust value function:

$$\widehat{V}_0^{\pi,\mathrm{rob}}(b_0) \triangleq \inf_{\substack{\widehat{P}^T_{0:H-1}\in\widehat{\mathcal{P}}^T_{0:H-1}\\ \widehat{P}^Z_{1:H}\in\widehat{\mathcal{P}}^Z_{1:H}}} \widehat{V}_0^{\pi}\Big(b_0;\, \widehat{P}^T_{0:H-1}, \widehat{P}^Z_{1:H}\Big)\,, \tag{37}$$

where the theorectical/projected value function are defined above (30), (31).

Using Lemma 3 together with the restriction maps $\mathcal{R}^T_{0:H-1}$ and $\mathcal{R}^Z_{1:H}$ from (28)-(29), that project the full uncertainty set onto the projected uncertainty set, we get the following equality:

$$\widehat{V}_0^{\pi,\mathrm{rob}}(b_0) = \inf_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \widehat{V}_0^{\pi}\big(b_0;\, \mathcal{R}^T_{0:H-1}(P^T_{0:H-1}), \mathcal{R}^Z_{1:H}(P^Z_{1:H})\big)\,. \tag{38}$$

The difference of the robust value function against the projected value function under non-rectangular setting can be bounded deterministically:

$$\begin{aligned}
&\Big|V_0^{\pi,\mathrm{rob}}(b_0) - \widehat{V}_0^{\pi,\mathrm{rob}}(b_0)\Big| \\
&= \left|\inf_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} V_0^{\pi}\big(b_0;\, P^T_{0:H-1}, P^Z_{1:H}\big) - \inf_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \widehat{V}_0^{\pi}\big(b_0;\, \mathcal{R}^T_{0:H-1}(P^T_{0:H-1}), \mathcal{R}^Z_{1:H}(P^Z_{1:H})\big)\right| &(39)\\
&\le \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \Big| V_0^{\pi}\big(b_0;\, P^T_{0:H-1}, P^Z_{1:H}\big) - \widehat{V}_0^{\pi}\big(b_0;\, \mathcal{R}^T_{0:H-1}(P^T_{0:H-1}), \mathcal{R}^Z_{1:H}(P^Z_{1:H})\big)\Big| &(40)
\end{aligned}$$

For the last transition we used Lemma 2.

To reduce clutter, we make use of the following shorthands:
i) $\Pr(\tau_{0:j}) = b_0(s_0)\prod_{k=0}^{j-1} P^T_k(s_{k+1}\mid\pi_k, s_k)P^Z_{k+1}(z_{k+1}\mid s_{k+1}) \rightarrow$ trajectory likelihood,
ii) The sampled trajectory set $\widehat{\mathcal{T}}_{1:j}$ defined in (7), and its complementary set $\mathcal{T}^{\mathrm{out}}_{1:j}$ defined in (27),
iii) $\mathcal{R}_{0:j}\Big(\widehat{\Pr}(\tau_{0:j})\Big) = \widehat{b}_0(s_0)\prod_{k=0}^{j-1}\mathcal{R}_{0:j}\Big(P^T_k(s_{k+1}|\pi_k, s_k)\Big)\mathcal{R}_{0:j}\Big(P^Z_{k+1}(z_{k+1}\mid s_{k+1})\Big)$,

$$= \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \left|\sum_{j=0}^{H-1}\left(\sum_{\tau_{0:j}\in\mathcal{T}_{0:j}} \Pr(\tau_{0:j})\; r(s_j,\pi_j)\right) - \left(\sum_{s_0} b_0(s_0)r(s_0,\pi_0) + \sum_{j=1}^{H-1}\sum_{\tau_{0:j}\in\mathcal{T}_{0:j}} \mathcal{R}_{0:j}\Big(\widehat{\Pr}(\tau_{0:j})\Big)\; r(s_j,\pi_j)\right)\right|, \tag{41}$$

at $j=0$, the root belief is the exact belief in both the theoretical and projected value, thus the immediate reward $r(b_0,\pi_0)$ is canceled.

$$= \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \left|\sum_{j=1}^{H-1}\left(\sum_{\tau_{0:j}\in\mathcal{T}_{0:j}} \Pr(\tau_{0:j})\, r(s_j,\pi_j) - \sum_{\tau_{0:j}\in\mathcal{T}_{0:j}} \mathcal{R}_{0:j}\Big(\widehat{\Pr}(\tau_{0:j})\Big)\; r(s_j,\pi_j)\right)\right|, \tag{42}$$

splitting the sums into in/out support:

$$= \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \left|\sum_{j=1}^{H-1}\left(\sum_{\tau_{0:j}\in\widehat{\mathcal{T}}_{0:j}} \underbrace{\left(\Pr(\tau_{0:j}) - \mathcal{R}_{0:j}\Big(\widehat{\Pr}(\tau_{0:j})\Big)\right)}_{\equiv 0}\; r(s_j,\pi_j) + \sum_{\tau^{\mathrm{out}}_{0:j}\in\mathcal{T}^{\mathrm{out}}_{0:j}}\left(\Pr(\tau^{\mathrm{out}}_{0:j}) - \underbrace{\mathcal{R}_{0:j}\big(\Pr(\tau^{\mathrm{out}}_{0:j})\big)}_{\text{=0, not defined outside}}\right) r(s_j,\pi_j)\right)\right|. \tag{43}$$

The underbraced terms in both summands equal zero by the definition of the Restricted mapping $\mathcal{R}$. Therefore: [don't start with = (throughout the document)]

$$= \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \left| \sum_{j=1}^{H-1} \sum_{\tau^{\text{out}}_{0:j}\in\mathcal{T}^{\text{out}}_{0:j}} \Pr(\tau^{\text{out}}_{0:j})\, r(s_j,\pi_j) \right| \leq R_{max} \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \sum_{j=1}^{H-1} \sum_{\tau^{\text{out}}_{0:j}\in\mathcal{T}^{\text{out}}_{0:j}} \Pr(\tau^{\text{out}}_{0:j}), \tag{44}$$

where $R_{max} \triangleq \max\big\{\mid r_{\min}\mid, \mid r_{\max}\mid\big\}$. Since $\Pr(\tau_{0:j})$ defines a full probability distribution over the full trajectory set $\mathcal{T}_{0:j}$, we have $\sum_{\tau_{0:j}\in\mathcal{T}_{0:j}} \Pr(\tau_{0:j}) \equiv 1$. Therefore:

$$\sum_{\tau^{\text{out}}_{0:j}\in\mathcal{T}^{\text{out}}_{0:j}} \Pr(\tau^{\text{out}}_{0:j}) = \Pr(\mathcal{T}^{\text{out}}_{0:j}) = 1 - \widehat{\Pr}(\widehat{\mathcal{T}}_{0:j}), \tag{45}$$

and (44) equals:

$$R_{max} \sup_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \left(\sum_{j=1}^{H-1} \Big(1 - \widehat{\Pr}(\widehat{\mathcal{T}}_{0:j})\Big) \right) = R_{max} \left(H - 1 - \inf_{\substack{P^T_{0:H-1}\in\mathcal{P}^T_{0:H-1}\\ P^Z_{1:H}\in\mathcal{P}^Z_{1:H}}} \sum_{j=1}^{H-1} \widehat{\Pr}(\widehat{\mathcal{T}}_{0:j}) \right). \tag{46}$$

Utilizing again Lemma 3 yields:

$$= R_{max} \left(H - 1 - \inf_{\substack{\widehat{P}^T_{0:H-1}\in\widehat{\mathcal{P}}^T_{0:H-1}\\ \widehat{P}^Z_{1:H}\in\widehat{\mathcal{P}}^Z_{1:H}}} \sum_{j=1}^{H-1} \sum_{\substack{z_{1:j}\in\widehat{\mathcal{Z}}(h_{1:j})\\ s_{0:j}\in\widehat{\mathcal{S}}(h_{0:j})}} \widehat{b}_0(s_0) \prod_{k=0}^{j-1} \widehat{P}^T_k(s_{k+1}\mid \pi_k, s_k)\, \widehat{P}^Z_{k+1}(z_{k+1}\mid s_{k+1}) \right) \triangleq \epsilon^\pi_0(b_0), \tag{47}$$

which concludes the proof. $\square$

Transition to a rectangular setting: If we assume rectangular uncertainty sets (5), the above bound (47) can be written as nested minimizations, that allows dynamic programming:

$$\begin{aligned}\epsilon^\pi_{0,rect}(b_0) \triangleq R_{max}\Bigg(H - 1 - &\sum_{s_0\in\widehat{\mathcal{S}}_0} b_0(s_0)\Bigg(\\ &\inf_{\substack{P^T_0\in\mathcal{P}^T_0\\ P^Z_1\in\mathcal{P}^Z_1}} \sum_{z_1\in\widehat{\mathcal{Z}}_1} \sum_{s_1\in\widehat{\mathcal{S}}_1} \widehat{P}^T_0(s_1\mid\pi_0,s_0)\, \widehat{P}^Z_1(z_1\mid s_1)\Bigg(\\ &1 + \inf_{\substack{P^T_1\in\mathcal{P}^T_1\\ P^Z_2\in\mathcal{P}^Z_2}} \sum_{z_2\in\widehat{\mathcal{Z}}_2} \sum_{s_2\in\widehat{\mathcal{S}}_2} \widehat{P}^T_1(s_2\mid\pi_1,s_1)\, \widehat{P}^Z_2(z_2\mid s_2)\Bigg(\cdots \\ &1 + \inf_{\substack{P^T_{H-2}\in\mathcal{P}^T_{H-2}\\ P^Z_{H-1}\in\mathcal{P}^Z_{H-1}}} \sum_{z_{H-1}\in\widehat{\mathcal{Z}}_{H-1}} \sum_{s_{H-1}\in\widehat{\mathcal{S}}_{H-1}} \widehat{P}^T_{H-2}(s_{H-1}\mid\pi_{H-2},s_{H-2})\, \widehat{P}^Z_{H-1}(z_{H-1}\mid s_{H-1})\Bigg)\cdots\Bigg)\end{aligned} \tag{48}$$

We can write (48) in a compact form, recursively:

$$\epsilon^\pi_{0,rect}(\widehat{b}_0) \triangleq R_{max}\left(H - 1 - \sum_{s_0\in\widehat{\mathcal{S}}_0} \widehat{b}_0(s_0)\delta^\pi_0(s_0)\right) \tag{49}$$

Such that for $t\in[0,H-2]$:

$$\delta^\pi_t(s_t) = \inf_{\substack{P^T_t\in\mathcal{P}^T_t\\ P^Z_{t+1}\in\mathcal{P}^Z_{t+1}}} \Bigg(\underbrace{\sum_{z_{t+1}\in\widehat{\mathcal{Z}}_{t+1}} \sum_{s_{t+1}\in\widehat{\mathcal{S}}_{t+1}} \widehat{P}^T_t(s_{t+1}\mid\pi_t,s_t)\, \widehat{P}^Z_{t+1}(z_{t+1}\mid s_{t+1})}_{=\widehat{\mathbb{E}}^{\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{z_{t+1}\mid\widehat{b}_t,a_t}} \left[1 + \delta^\pi_{t+1}(s_{t+1})\right]\Bigg) \tag{50}$$

and for $t = H-1$ there is no continuation therefore:

$$\delta^\pi_{H-1}(s_{H-1}) \equiv 0 \tag{51}$$

Proof of Lemma 1

Lemma 1. *Let H be the horizon of the POMDP, and π be the policy. the theoretical/projected value function V_t^π, $\widehat{V}_t^\pi$ at time step t equals:*

$$V_t^\pi(b_t) = \sum_{j=t}^{H-1} \sum_{z_{t+1:j}} \sum_{s_{t:j}} b_t(s_t) \prod_{k=t}^{j-1} P_k^T(s_{k+1} \mid s_k, \pi_k) P_{k+1}^Z(z_{k+1} \mid s_{k+1}) r\big(s_j, \pi_j(h_j)\big), \tag{52}$$

$$\widehat{V}_t^\pi(b_t) = \sum_{s_t} b_t(s_t) r\big(s_t, \pi_t(h_t)\big) + \sum_{j=t+1}^{H-1} \sum_{\substack{z_{t+1:j} \in \widehat{\mathcal{Z}}_{t+1:j} \\ s_{t:j} \in \widehat{\mathcal{S}}_{t:j}}} \widehat{b}_t(s_t) \prod_{k=t}^{j-1} \widehat{P}_k^T(s_{k+1}|s_k, \pi_k) \widehat{P}_{k+1}^Z(z_{k+1}|s_{k+1}) r\big(s_j, \pi_j(h_j)\big). \tag{53}$$

Proof.
The belief dependent reward by definition is:

$$r(b_t(h_t), a_t) = \mathrm{E}_{s_t \sim b_t}[r(s_t, a_t)] = \sum_{s_t} b_t(s_t) r(s_t, a_t). \tag{54}$$

The value function is the expected future cumulative reward:

$$V_t^\pi(b_t) = \mathbb{E}_{z_{t+1:j}}^\pi \left[\sum_{j=t}^{H-1} r(b_j, \pi_j) | b_t \right] = \sum_{j=t}^{H-1} \mathbb{E}_{z_{t+1:j}}^\pi \left[r(b_j, \pi_j) | b_t \right]. \tag{55}$$

b_j is the belief obtained after starting from b_t and observing the sequence $z_{t+1:j}$. Using (54) and (55):

$$V_t^\pi(b_t) = \sum_{j=t}^{H-1} \sum_{z_{t+1:j}} \sum_{s_j} P(z_{t+1:j} \mid b_t) b_j(s_j) r(s_j, \pi_j). \tag{56}$$

Applying chain rule we rewrite $P(z_{t+1:j} \mid b_t) b_j(s_j) = P(s_j, z_{t+1:j} \mid b_t)$ and marginalizing over the hidden states sequence:

$$P(s_j, z_{t+1:j} \mid b_t) = \sum_{s_{t:j-1}} P(s_{t:j}, z_{t+1:j} \mid b_t). \tag{57}$$

Assuming Markovian models, the joint probability of the state-observation trajectory can be written as:

$$P(s_{t:j}, z_{t+1:j} \mid b_t) = b_t(s_t) \prod_{k=t}^{j-1} P_k^T(s_{k+1} \mid s_k, a_k) P_{k+1}^Z(z_{k+1} \mid s_{k+1}). \tag{58}$$

Plugging (57), (58) into value function (55) gives the final expression for the value function:

$$V_t^\pi(b_t) = \sum_{j=t}^{H-1} \sum_{z_{t+1:j}} \sum_{s_{t:j}} b_t(s_t) \prod_{k=t}^{j-1} P_k^T(s_{k+1} \mid s_k, \pi_k) P_{k+1}^Z(z_{k+1} \mid s_{k+1}) r(s_j, \pi_j). \tag{59}$$

Similarly, using the projected models defined in (8), (9), and the restricted belief (10), the simplified value function is:

$$\widehat{V}_t^\pi(b_t) = \sum_{s_t} b_t(s_t) r\big(s_t, \pi_t(h_t)\big) + \sum_{j=t+1}^{H-1} \sum_{\substack{z_{t+1:j} \in \widehat{\mathcal{Z}}_{t+1:j} \\ s_{t:j} \in \widehat{\mathcal{S}}_{t:j}}} \widehat{b}_t(s_t) \prod_{k=t}^{j-1} \widehat{P}_k^T(s_{k+1}|s_k, \pi_k) \widehat{P}_{k+1}^Z(z_{k+1}|s_{k+1}) r\big(s_j, \pi_j(h_j)\big), \tag{60}$$

where the unnormalized projected belief $\widehat{b}_t$ is updated using the projected models, namely:

$$\widehat{b}_t(s_t; \widehat{\mathcal{S}}(h_t)) = \widehat{P}_t^Z(z_t|s_t) \sum_{s_{t-1} \in \widehat{\mathcal{S}}(h_{t-1}^-)} \widehat{P}_{t-1}^T(s_t|s_{t-1}, \pi_{t-1}) \widehat{b}(s_{t-1}) \tag{61}$$

$\square$

Proof of Lemma 2

Lemma 2. *for any 2 function $f(\theta), \widehat{f}(\theta)$, the following holds:*

$$\left|\inf_{\theta} f(\theta) - \inf_{\theta} \hat{f}(\theta)\right| \leq \sup_{\theta} \left|f(\theta) - \hat{f}(\theta)\right| \tag{62}$$

Proof.
Let

$$\varepsilon = \sup_{\theta \in \Theta} |f(\theta) - g(\theta)|$$

Then for every θ,

$$f(\theta) \leq g(\theta) + \varepsilon$$

Taking minima gives

$$\inf_{\theta} f(\theta) \leq \inf_{\theta} g(\theta) + \varepsilon$$

Similarly, since

$$g(\theta) \leq f(\theta) + \varepsilon,$$

we get

$$\inf_{\theta} g(\theta) \leq \inf_{\theta} f(\theta) + \varepsilon$$

Therefore,

$$\left|\inf_{\theta} f(\theta) - \inf_{\theta} g(\theta)\right| \leq \varepsilon = \sup_{\theta \in \Theta} |f(\theta) - g(\theta)|$$

□

Proof of Lemma 3

Lemma 3. *Let $\mathcal{P}$ denote some uncertainty set, and $\widehat{\mathcal{P}}$ denote its projected version. Let $\mathcal{R}$ be the restriction map that takes a full probability distribution $P \in \mathcal{P}$ and retains only the entries associated with the sampled support:*

$$\mathcal{R}(P; \widehat{\mathcal{S}}) = \begin{cases} P(s), & s \in \widehat{\mathcal{S}}, \\ 0, & \textit{otherwise}. \end{cases} \tag{63}$$

Then minimizing some function $f : \widehat{P} \to \mathbb{R}$ over $\widehat{\mathcal{P}}$ is the same as minimizing $f(\mathcal{R}(P))$ over the full set $\mathcal{P}$, namely:

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) = \inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}), \tag{64}$$

Proof.
By the definition of the projected uncertainty sets as the images of the full uncertainty sets under the restriction maps, we have

$$\widehat{\mathcal{P}} = \mathcal{R}\left(\mathcal{P}\right), \tag{65}$$

We prove the desired equality by showing both inequalities. First, let

$$P \in \mathcal{P}, \tag{66}$$

be arbitrary full distribution. By definition:

$$\mathcal{R}(P) \in \widehat{\mathcal{P}}, \tag{67}$$

Therefore,

$$\inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}) \leq f(\mathcal{R}(P)). \tag{68}$$

Since this holds for every

$$P \in \mathcal{P}, \tag{69}$$

we can take the infimum over the full uncertainty set and obtain

$$\inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}) \leq \inf_{P \in \mathcal{P}} f(\mathcal{R}(P)). \tag{70}$$

For the opposite direction, let

$$\widehat{P} \in \widehat{\mathcal{P}}, \tag{71}$$

be arbitrary projected distribution. Since the projected uncertainty set is defined as the image of the full uncertainty set, there exist a full model:

$$P \in \mathcal{P}, \tag{72}$$

such that

$$\widehat{P} = \mathcal{R}(P), \tag{73}$$

Therefore,

$$f(\widehat{P}) = f(\mathcal{R}(P)) \tag{74}$$

Hence,

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) \leq f(\widehat{P}). \tag{75}$$

Since this holds for every

$$\widehat{P} \in \widehat{\mathcal{P}}, \tag{76}$$

we can take the infimum over the projected uncertainty set and obtain

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) \leq \inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}). \tag{77}$$

Combining the two inequalities gives equality:

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) = \inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}). \tag{78}$$

Which proves the Lemma.

□

Proof of Corollary 1

Corollary 1 (restated). *The bound of Theorem 1 holds for every node h_t of the tree, given that along the trajectory to h_t ,each previous node's belief was updated over the full state space (exact belief updates), namely:*

$$\left| V_t^{\pi,\mathrm{rob}}(b_t) - \widehat{V}_t^{\pi,\mathrm{rob}}(b_t) \right| \leq \epsilon_t^{\pi}(b_t) \tag{79}$$

$$\epsilon_t^{\pi}(b_t) \triangleq R_{max} \left(H - t - 1 - \inf_{\substack{\widehat{P}^T_{t:H-1} \in \widehat{\mathcal{P}}^T_{t:H-1} \\ \widehat{P}^Z_{t+1:H} \in \widehat{\mathcal{P}}^Z_{t+1:H}}} \sum_{j=t+1}^{H-1} \sum_{z_{t+1:j} \in \widehat{\mathcal{Z}}_{t+1:j}} \sum_{s_{t:j} \in \widehat{\mathcal{S}}_{t:j}} b_t(s_t) \prod_{k=t}^{j-1} \widehat{P}^T_k(s_{k+1} \mid \pi_k, s_k)\, \widehat{P}^Z_{k+1}(z_{k+1} \mid s_{k+1}) \right) \tag{80}$$

Proof.
The proof of theorem 1 relies on normalized belief in (45), namely:

$$\sum_{\tau_{t:j} \in \mathcal{T}_{t:j}} \Pr(\tau_{t:j}) = \sum_{z_{t+1:j}} \sum_{s_{t:j}} b_t(s_t) \prod_{k=t}^{j-1} P^T_k(s_{k+1} \mid \pi_k, s_k)\, P^Z_{k+1}(z_{k+1} \mid s_{k+1}) \equiv 1, \tag{81}$$

which requires only that the belief at the node be a true normalized distribution. Under exact belief updates this holds at every history node h_t and the theorem applies precisely with b_0, t=0 replaced by b_t, t.

□

NOTE: the bound is still a function of the projected models $P^T_{t:H-1}, P^Z_{t+1:H}$ resulting in an optimization over a much lower dimensionality! We thus pay additional computation only in the belief-update, while the optimizations remain cheap.

Proof of Corollary 2

Corollary 2 (restated). *For any policy* π*, as the sampled trajectories set increases* $\widehat{\mathcal{T}} \to \mathcal{T}$ *the projected robust value function* (14) *convergences to the theoretical robust value function* (3)*:*

$$\widehat{V}_t^{\pi,\mathrm{rob}}(\widehat{b}_t) \xrightarrow{\widehat{\mathcal{T}} \to \mathcal{T}} V_t^{\pi,\mathrm{rob}}(b_t). \tag{82}$$

Proof. ADD PROOF:
general idea:

$$\left| V_0^{\pi,\mathrm{rob}}(b_0) - \widehat{V}_0^{\pi,\mathrm{rob}}(b_0) \right| \;\leq\; \epsilon_0^{\pi}(b_0) \tag{83}$$

$\epsilon_0^{\pi}(b_0)$ shrinks to zero as support increases. □

Proof of Theorem 2:

Essentials lemmas and definitions for the proof:

Lemma 4 (restated). *Let b_t be the nominal-belief at time step t, and b_{t+1} be the nominal-belief at the following time step $t+1$. For any action $a_t \in \mathcal{A}_t$, and future action a_{t+1} selected by the RUDB-greedy policy $\pi_{t+1}(b_{t+1}) = \underset{a_{t+1}\in\mathcal{A}_{t+1}}{\arg\max}\left\{RUDB^{\pi}(b_{t+1}, a_{t+1})\right\}$, the following inequality holds:*

$$r(b_t, a_t) + \inf_{\substack{P_t^T\in\mathcal{P}_t^T\\ P_{t+1}^Z\in\mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}[RUDB^{\pi}(b_{t+1})] \leq RUDB^{\pi}(b_t, a_t).$$

We restate for convenience the definitions of $RUDB^{\pi}(b_t, a_t)$ (20), and its derivative terms:

$$RUDB^{\pi}(b_t, a_t) \triangleq r(b_t, a_t) + \inf_{\substack{\widehat{P}_t^T\in\widehat{\mathcal{P}}_t^T\\ \widehat{P}_{t+1}^Z\in\widehat{\mathcal{P}}_{t+1}^Z}} \left\{\widehat{\mathbb{E}}_{z_{t+1}|b_t,a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right] + \epsilon_{t,rect}^{\pi,\widehat{P}_t^T,\widehat{P}_{t+1}^Z}(b_t, a_t)\right\}, \tag{84}$$

$$= \widehat{Q}_{t,rect}^{\pi,RUDB}(b_t, a_t) + \epsilon_{t,rect}^{\pi,RUDB}(b_t, a_t). \tag{85}$$

IMPORTANT: the RUDB superscript denotes the **shared** optimized models $\widehat{P}_t^{*T}, \widehat{P}_{t+1}^{*Z}$ computed by the infimum, meaning:

$$\widehat{P}_t^{*T}, \widehat{P}_{t+1}^{*Z} = arg \inf_{\substack{\widehat{P}_t^T\in\widehat{\mathcal{P}}_t^T\\ \widehat{P}_{t+1}^Z\in\widehat{\mathcal{P}}_{t+1}^Z}} \left\{\widehat{\mathbb{E}}_{z_{t+1}|b_t,a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right] + \epsilon_{t,rect}^{\pi,\widehat{P}_t^T,\widehat{P}_{t+1}^Z}(b_t, a_t)\right\}, \tag{86}$$

$$\widehat{Q}_{t,rect}^{\pi,RUDB}(b_t, a_t) \triangleq r(b_t, a_t) + \widehat{\mathbb{E}}_{z_{t+1}|b_t,a_t}^{\widehat{P}_t^{*T}, \widehat{P}_{t+1}^{*Z}}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right], \tag{87}$$

$$\epsilon_{t,rect}^{\pi,\widehat{P}_t^{*T},\widehat{P}_{t+1}^{*Z}}(b_t, a_t) \triangleq R_{max}\left(H - t - 1 - \sum_{s_t\in\widehat{\mathcal{S}}_t} b_t(s_t)\delta_t^{\pi,\widehat{P}_t^{*T},\widehat{P}_{t+1}^{*Z}}(s_t, a_t)\right), \tag{88}$$

$$\delta_t^{\pi,\widehat{P}_t^{*T},\widehat{P}_{t+1}^{*Z}}(s_t, a_t) = \sum_{z_{t+1}\in\widehat{\mathcal{Z}}_{t+1}} \sum_{s_{t+1}\in\widehat{\mathcal{S}}_{t+1}} \widehat{P}_t^{*T}(s_{t+1}|a_t, s_t)\, \widehat{P}_{t+1}^{*Z}(z_{t+1}|s_{t+1}) \left[1 + \delta_{t+1}^{\pi}(s_{t+1})\right], \tag{89}$$

$$b(s_{t+1}) = \frac{\widehat{P}_{t+1}^{Z}(z_{t+1}|s_{t+1}) \cdot \sum_{s_t\in\mathcal{S}} \mathring{P}_t^T(s_{t+1}|s_t, a_t)b(s_t)}{\Pr(z_{t+1}|b_t, a_t)} \rightarrow \textbf{exact belief updates}. \tag{90}$$

Theorem 2 (restated). *Let b_t be the belief at time step t, $\mathring{P}_t^T$ be the nominal transition model [what about the observation model?] that is used to update the posterior belief $b_{t+1}(s_{t+1}|h_t^-, z_{t+1}; \mathring{P}_t^T) = \Pr(s_{t+1}|h_t)$, called the nominal-belief at time step $t+1$. The optimal robust value function at time step t over rectangular uncertainty sets* (5)*, is upper bounded by:*

$$V_{t,rect}^{*,\text{rob}}(b_t) \leq RUDB^{\pi}(b_t) \triangleq \max_{a_t\in\mathcal{A}}\left\{RUDB^{\pi}(b_t, a_t)\right\}, \tag{91}$$

such that the policy π comes from applying Bellman optimality operator over $RUDB^{\pi}(b_t, a_t)$:

$$\pi_t(b_t) = \underset{a_t\in\mathcal{A}_t}{\arg\max}\left\{\widehat{Q}_{t,rect}^{\pi,RUDB}(b_t, a_t) + \epsilon_{t,rect}^{\pi,RUDB}(b_t, a_t)\right\}, \tag{92}$$

Proof. We prove by induction, that by applying the Bellman optimality operator on upper bounds to the robust value function, we get an upper bound on the **optimal** value function.

Base case ($t = H - 1$): We set the terminal action-value to be the immediate expected reward:

$$Q_{H,rect}^{*,\text{rob}}(b_{H-1}, a_{H-1}) = \widehat{Q}_{H-1,rect}^{\pi,\text{rob}}(b_{H-1}, a_{H-1}) = r(b_{H-1}, a_{H-1}) = \sum_{s_{H-1}\in S} b_{H-1}(s_{H-1})r(s_{H-1}, a_{H-1}). \tag{93}$$

At the terminal time step $H-1$ of the POMDP, there is no future transition/observation term, therefore $\epsilon^{\pi}_{H-1,rect}(b_{H-1},a_{H-1})=0$. Therefore the terminal $RUDB^{\pi}(b_{H-1},a_{H-1})$ on the action-value is:

$$RUDB^{\pi}(b_{H-1},a_{H-1})=r(b_{H-1},a_{H-1}) \tag{94}$$

and the base-case for the induction is the best immediate expected reward:

$$V^{*,\text{rob}}_{H-1,rect}(b_{H-1}) \leq RUDB^{\pi}(b_{H-1}) \triangleq \max_{a_{H-1}\in\mathcal{A}_{H-1}} RUDB^{\pi}(b_{H-1},a_{H-1}). \tag{95}$$

Induction Hypothesis: Given time step $t+1$, assume that for any future sampled-observation $z_{t+1}\in\widehat{\mathcal{Z}}_{t+1}(h_{t+1})$ and any nominal-belief $b_{t+1}(s_{t+1}|h^{-}_{t},z_{t+1};\mathring{P}^{T}_{t})$:

$$V^{*,\text{rob}}_{t+1,rect}(b_{t+1}) \leq RUDB^{\pi}(b_{t+1}). \tag{96}$$

This means that at the next depth of the tree $RUDB^{\pi}(b_{t+1})$ is already a valid upper bound on the optimal rectangular robust value.

Induction Step: We will now show that the hypothesis holds for time step t, specifically:

$$Q^{*,\text{rob}}_{t,rect}(b_t,a_t) \leq RUDB^{\pi}(b_t,a_t). \tag{97}$$

Writing the above LHS recursively:

$$Q^{*,\text{rob}}_{t,\text{rect}}(b_t,a_t) = r(b_t,a_t) + \inf_{\substack{P^T_t\in\mathcal{P}^T_t \\ P^Z_{t+1}\in\mathcal{P}^Z_{t+1}}} \mathbb{E}^{P^T_t,P^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[V^{*,\text{rob}}_{t+1,\text{rect}}(b_{t+1})\right], \tag{98}$$

by the induction hypothesis for any nominal-belief b_{t+1} at time step $t+1$:

$$\leq r(b_t,a_t) + \inf_{\substack{P^T_t\in\mathcal{P}^T_t \\ P^Z_{t+1}\in\mathcal{P}^Z_{t+1}}} \mathbb{E}^{P^T_t,P^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[RUDB^{\pi}(b_{t+1})\right],$$

following Lemma 4:

$$\leq \widehat{Q}^{\pi,RUDB}_{t,rect}(b_t,a_t) + \epsilon^{\pi,RUDB}_{t,rect}(b_t,a_t) \triangleq RUDB^{\pi}(b_t,a_t), \tag{99}$$

since:

$$V^{*,rob}_{t,rect}(b_t) = \max_{a_t\in\mathcal{A}_t} Q^{*,rob}_{t,rect}(b_t,a_t) \leq \max_{a_t\in\mathcal{A}_t} RUDB^{\pi}(b_t,a_t) \triangleq RUDB^{\pi}(b_t), \tag{100}$$

we proved the induction step, and the theorem.

□

Proof of Lemma 4

Lemma 4. *Let b_t be the nominal-belief at time step t, and b_{t+1} be the nominal-belief at the following time step $t+1$. For any action $a_t \in \mathcal{A}_t$, and future action a_{t+1} selected by the RUDB-greedy policy $\pi_{t+1}(b_{t+1}) = \underset{a_{t+1}\in\mathcal{A}_{t+1}}{\arg\max}\Big\{RUDB^{\pi}(b_{t+1}, a_{t+1})\Big\}$, the following inequality holds:*

$$r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}[RUDB^{\pi}(b_{t+1})] \leq RUDB^{\pi}(b_t, a_t).$$

Proof.
By the definition of the $RUDB^{\pi}(b_{t+1}, a_{t+1})$ (20), at time step $t+1$:

$$\begin{aligned}
&r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}[RUDB^{\pi}(b_{t+1})] \\
=&r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\max_{a_{t+1}\in\mathcal{A}_t}\Big\{\widehat{Q}_{t+1,rect}^{\pi,RUDB}(b_{t+1}, a_{t+1}) + \epsilon_{t+1,rect}^{\pi,RUDB}(b_{t+1}, a_{t+1})\Big\}\right]. \quad (101)
\end{aligned}$$

The action a_{t+1} is selected after observing z_{t+1}, meaning after reaching the next history $h_{t+1} = (h_t, a_t, z_{t+1})$. Therefore, for each posterior nominal-belief $b_{t+1|z_{t+1}}$, there is a corresponding action that maximizes (101):

$$a_{t+1}^{RUDB} = \pi_{t+1}(b_{t+1}|z_{t+1}) = \underset{a_{t+1}\in\mathcal{A}_t}{\arg\max}\{\widehat{Q}_{t+1,rect}^{\pi,RUDB}(b_{t+1}, a_{t+1}) + \epsilon_{t+1,rect}^{\pi,RUDB}(b_{t+1}, a_{t+1})\} \quad (102)$$

$$= r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}) + \epsilon_{t+1,rect}^{\pi,RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1}))\right], \quad (103)$$

note that we used: $\widehat{Q}_{t+1,rect}^{\pi,RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \to \widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})$.

By the linearity of the expectation operator $\mathbb{E}[\cdot]$:

$$= r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \Big\{\mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right] + \mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\epsilon_{t+1,rect}^{\pi,RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1}))\right]\Big\}. \quad (104)$$

We first focus on the **left** expectation from (104), and show that $\mathbb{E}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}\right] = \widehat{\mathbb{E}}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}\right]$ [this can be confusing, suggest to show explicitly, i.e. $\mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right]$ and the corresponding expression for $\widehat{\mathbb{E}}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right]$].
$\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})$ is a function of the the belief b_{t+1}, hence a function of observation z_{t+1}. Therefore:

$$\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}) \equiv 0, \quad \forall z_{t+1} \notin \widehat{\mathcal{Z}}(h_{t+1}), \quad (105)$$

writing the expectation $\mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right]$ explicitly:

$$\mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z}\left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})\right] = \sum_{z_{t+1}\in Z_{t+1}} \Pr(z_{t+1}|b_t, a_t)\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}) \quad (106)$$

we split the sums to in/out of sampled-support:

$$\begin{aligned}
&= \sum_{z_{t+1}\in\widehat{\mathcal{Z}}(h_{t+1})} \Pr(z_{t+1}|b_t, a_t)\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}) + \sum_{z_{t+1}\notin\widehat{\mathcal{Z}}(h_{t+1})} \Pr(z_{t+1}|b_t, a_t)\underbrace{\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1})}_{=0\text{ off support eq (105)}}, \quad (107)\\
&= \sum_{z_{t+1}\in\widehat{\mathcal{Z}}(h_{t+1})} \Pr(z_{t+1}|b_t, a_t)\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}), \quad (108)
\end{aligned}$$

note that on the sampled support $z_{t+1} \in \widehat{\mathcal{Z}}(h_{t+1})$: $\Pr(z_{t+1}|b_t, a_t) = \widehat{\Pr}(z_{t+1}|b_t, a_t)$, since:

$$\Pr(z_{t+1}|b_t, a_t) = \sum_{s_t} b_t(s_t) \sum_{s_{t+1}} P_t^T(s_{t+1}|s_t, \pi_t) P_{t+1}^Z(z_{t+1}|s_{t+1}), \tag{109}$$

$$= \sum_{s_t} b_t(s_t) \left(\sum_{s_{t+1} \in \widehat{\mathcal{S}}(h_{t+1})} P_t^T(s_{t+1}|s_t, \pi_t) + \underbrace{\sum_{s_{t+1} \notin \widehat{\mathcal{S}}(h_{t+1})} P_t^T(s_{t+1}|s_t, \pi_t)}_{=0} \right) P_{t+1}^Z(z_{t+1}|s_{t+1}), \tag{110}$$

the underbraced term vanishes since we are on support $z_{t+1} \in \widehat{\mathcal{Z}}(h_{t+1})$ which is a function of the history h_{t+1}, thus h_{t+1}^- induces $\widehat{\mathcal{S}}(h_{t+1}^-)$. On the sampled support the models P_t^T, P_{t+1}^Z coincide with their projected counterparts $\widehat{P}_t^T, \widehat{P}_{t+1}^Z$:

$$= \sum_{s_t} b_t(s_t) \sum_{s_{t+1} \in \widehat{\mathcal{S}}(h_{t+1})} \widehat{P}_t^T(s_{t+1}|s_t, \pi_t) \widehat{P}_{t+1}^Z(z_{t+1}|s_{t+1}), \tag{111}$$

therefore (106) equals:

$$\sum_{z_{t+1} \in \widehat{\mathcal{Z}}_{t+1}} \widehat{\Pr}(z_{t+1} \mid b_t, a_t) \widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}) \; = \; \widehat{\mathbb{E}}_{z_{t+1}|b_t,a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z} \left[\widehat{V}_{t+1,rect}^{\pi,RUDB}(b_{t+1}) \right] \tag{112}$$

Moving on to the **right** expectation in (104) we now show that: $\mathbb{E}\left[\epsilon_{t+1,rect}^{\pi,RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1}))\right] \leq \epsilon_{t,rect}^{\pi,\widehat{P}_t^T,\widehat{P}_{t+1}^Z}(b_t, a_t)$:

$$\begin{aligned}
\mathbb{E}_{z_{t+1}|b_t,a_t}^{P_t^T, P_{t+1}^Z} \left[\epsilon_{t+1,rect}^{\pi,RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \right] &= \sum_{z_{t+1}} \Pr(z_{t+1}|b_t, a_t) R_{max} \left(H - t - 2 - \sum_{s_{t+1}} b_{t+1}(s_{t+1}) \delta_{t+1}^{\pi}(s_{t+1}, \pi_{t+1}) \right) \\
&= R_{max} \left(H - t - 2 - \sum_{z_{t+1}} \Pr(z_{t+1}|b_t, a_t) \sum_{s_{t+1}} b_{t+1}(s_{t+1}) \delta_{t+1}^{\pi}(s_{t+1}, \pi_{t+1}) \right).
\end{aligned} \tag{113}$$

The last transition is due to $\sum_{z_{t+1}} \Pr(z_{t+1}|b_t, a_t)(H - t - 2) = (H - t - 2)$. The posterior belief $b_{t+1}(s_{t+1})$ by definition of Bayesian updates:

$$b(s_{t+1}) = \frac{\widehat{P}_{t+1}^Z(z_{t+1}|s_{t+1}) \cdot \sum_{s_t \in \mathcal{S}} \mathring{P}_t^T(s_{t+1}|s_t, a_t) b(s_t)}{\Pr(z_{t+1}|b_t, a_t)}, \tag{114}$$

therefore the denominator cancels out:

$$= R_{max} \left(H - t - 2 - \sum_{s_t} b(s_t) \sum_{z_{t+1}} \sum_{s_{t+1}} \widehat{P}_{t+1}^Z(z_{t+1}|s_{t+1}) \mathring{P}_t^T(s_{t+1}|s_t, a_t) \delta_{t+1}^{\pi}(s_{t+1}, \pi_{t+1}) \right), \tag{115}$$

we can use the projected nominal transition model $\widehat{P}_t^T(s_{t+1}|s_t, a_t) = \mathcal{R}_t^T\left(\mathring{P}_t^T(s_{t+1}|s_t, a_t)\right)$, to get the following inequality (sums iterating only over a subset):

$$\leq R_{max}\left(H - t - 2 - \sum_{s_t} b(s_t) \sum_{\substack{z_{t+1}\widehat{\mathcal{Z}}(h_{t+1}) \\ s_{t+1}\widehat{\mathcal{S}}(h^-_{t+1})}} \widehat{P}^Z_{t+1}(z_{t+1}|s_{t+1})\widehat{P}^T_t(s_{t+1}|s_t, a_t)\delta^\pi_{t+1}(s_{t+1}, \pi_{t+1})\right), \tag{116}$$

$$= R_{max}\left(H - t - 1 - 1 - \sum_{s_t} b(s_t) \sum_{\substack{z_{t+1}\widehat{\mathcal{Z}}(h_{t+1}) \\ s_{t+1}\widehat{\mathcal{S}}(h^-_{t+1})}} \widehat{P}^Z_{t+1}(z_{t+1}|s_{t+1})\widehat{P}^T_t(s_{t+1}|s_t, a_t)\delta^\pi_{t+1}(s_{t+1}, \pi_{t+1})\right), \tag{117}$$

$$\leq R_{max}\left(H - t - 1 - \left(\sum_{s_t} b(s_t) \sum_{\substack{z_{t+1}\widehat{\mathcal{Z}}(h_{t+1}) \\ s_{t+1}\widehat{\mathcal{S}}(h^-_{t+1})}} \widehat{P}^Z_{t+1}(z_{t+1}|s_{t+1})\widehat{P}^T_t(s_{t+1}|s_t, a_t)\right)\right. \tag{118}$$

$$\left. - \sum_{s_t} b(s_t) \sum_{\substack{z_{t+1}\widehat{\mathcal{Z}}(h_{t+1}) \\ s_{t+1}\widehat{\mathcal{S}}(h^-_{t+1})}} \widehat{P}^Z_{t+1}(z_{t+1}|s_{t+1})\widehat{P}^T_t(s_{t+1}|s_t, a_t)\delta^\pi_{t+1}(s_{t+1}, \pi_{t+1})\right), \tag{119}$$

$$= R_{max}\left(H - t - 1 - \sum_{s_t} b(s_t) \underbrace{\sum_{\substack{z_{t+1}\widehat{\mathcal{Z}}(h_{t+1}) \\ s_{t+1}\widehat{\mathcal{S}}(h^-_{t+1})}} \widehat{P}^Z_{t+1}(z_{t+1}|s_{t+1})\widehat{P}^T_t(s_{t+1}|s_t, a_t)\left[\delta^\pi_{t+1}(s_{t+1}, \pi_{t+1}) + 1\right]\right)}_{\delta_t^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}(s_t,a_t)}, \tag{120}$$

$$= R_{max}\left(H - t - 1 - \sum_{s_t} b(s_t)\delta_t^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}(s_t, a_t)\right) \triangleq \epsilon^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{t,rect}(b_t, a_t). \tag{121}$$

$\epsilon^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{t,rect}(b_t, a_t)$ denotes the error for fixed current models $\widehat{P}^T_t, \widehat{P}^Z_{t+1}$. Referring back to (104), we have shown that:

$$= r(b_t, a_t) + \inf_{\substack{P^T_t \in \mathcal{P}^T_t \\ P^Z_{t+1} \in \mathcal{P}^Z_{t+1}}} \left\{\mathbb{E}^{P^T_t,P^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[\widehat{V}^{\pi,RUDB}_{t+1,rect}(b_{t+1})\right] + \mathbb{E}^{P^T_t,P^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[\epsilon^{\pi,RUDB}_{t+1,rect}(b_{t+1}, \pi_{t+1}(b_{t+1}))\right]\right\} \tag{122}$$

$$\leq r(b_t, a_t) + \inf_{\substack{P^T_t \in \mathcal{P}^T_t \\ P^Z_{t+1} \in \mathcal{P}^Z_{t+1}}} \left\{\widehat{\mathbb{E}}^{\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[\widehat{V}^{\pi,RUDB}_{t+1,rect}(b_{t+1})\right] + \epsilon^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{t,rect}(b_t, a_t)\right\}. \tag{123}$$

We use Lemma 3, which shows that the last infimum can be over the projected uncertainty sets, namely:

$$\inf_{P \in \mathcal{P}} \widehat{f}(\mathcal{R}(P)) = \inf_{\widehat{P} \in \widehat{\mathcal{P}}} \widehat{f}(\widehat{P}), \tag{124}$$

Therefore

$$= r(b_t, a_t) + \inf_{\substack{\widehat{P}^T_t \in \widehat{\mathcal{P}}^T_t \\ \widehat{P}^Z_{t+1} \in \widehat{\mathcal{P}}^Z_{t+1}}} \left\{\widehat{\mathbb{E}}^{\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{z_{t+1}|b_t,a_t}\left[\widehat{V}^{\pi,RUDB}_{t+1,rect}(b_{t+1})\right] + \epsilon^{\pi,\widehat{P}^T_t,\widehat{P}^Z_{t+1}}_{t,rect}(b_t, a_t)\right\} \tag{125}$$

$$\triangleq RUDB^\pi(b_t, a_t), \tag{126}$$

which concludes the proof. □

Proof of Corollary 3

Corollary 3 (restated). *Let b_t be the belief as time step t, and π^{RUDB} be the RUDB policy* (24). *As the sampled trajectories set increases $\widehat{\mathcal{T}} \to \mathcal{T}$, the projected robust value function [need to define it somewhere] convergences to the optimal robust value function over rectangular uncertainty sets, namely:*

$$\widehat{V}^{\pi^{RUDB},\mathrm{rob}}_{t,rect}(b_t) \xrightarrow{\widehat{\mathcal{T}} \to \mathcal{T}} V^{*,\mathrm{rob}}_{t,rect}(b_t) \tag{127}$$

ADD PROOF:
general idea:

$$\widehat{V}_t^{\pi,\mathrm{rob}}(b_t) - \epsilon_t^{\pi}(b_t) \leq V_t^{\pi,\mathrm{rob}}(b_t) \leq \widehat{V}_t^{\pi,RUDB}(b_t) + \epsilon_t^{\pi,RUDB}(b_t) \tag{128}$$

Both $\epsilon_t^{\pi}(b_t), \epsilon_t^{\pi,RUDB}(b_t)$ shrink to zero as support increases. as that happen $\widehat{V}_t^{\pi,RUDB}(b_t) \to \widehat{V}_t^{\pi,\mathrm{rob}}(b_t)$